Purine Metabolism and its Disorders / Gout

Committed rate-limiting step: PRPP amidotransferase synthesizes 5-phosphoribosyl-1-amine using glutamine, allosterically inhibited by AMP, GMP, and IMP, and activated by PRPP.

Salvage pathway enzyme: HGPRT salvages hypoxanthine to IMP and guanine to GMP using PRPP; its deficiency results in **Lesch-Nyhan syndrome** (X-linked recessive, presenting with hyperuricemia, self-mutilation, dystonia, and orange urate sand in diapers).

Gout presentation and microscopy: Deposition of monosodium urate crystals displaying needle-shaped morphology and **strong negative birefringence** (yellow when parallel to slow vibration axis under polarized light).

Targeted therapy: Allopurinol is a suicide inhibitor of xanthine oxidase, metabolized to oxypurinol/alloxanthine, shifting purine excretion to the far more water-soluble hypoxanthine and xanthine.

Severe Combined Immunodeficiency (SCID): Caused by **adenosine deaminase (ADA)** deficiency, leading to toxic accumulation of dATP, which allosterically arrests ribonucleotide reductase and blocks DNA synthesis in T and B lymphocytes.

DNA Replication and Disorders

- **Replication polarity:** DNA synthesis proceeds exclusively **5' to 3'**, reading the parent template 3' to 5'; the leading strand is continuous, whereas the lagging strand forms Okazaki fragments.

- **RNA primer synthesis:** Initiated by primase (prokaryotes) or DNA polymerase (eukaryotes), which synthesizes a short RNA primer to provide the free 3'-OH required by replicative DNA polymerases.
- **Prokaryotic vs. eukaryotic polymerases:** Bacterial DNA Pol III handles bulk replication (possesses 3' to 5' proofreading exonuclease), DNA Pol I removes primers via its unique **5' to 3' exonuclease activity**; eukaryotic DNA Pol replicates the lagging strand and DNA Pol replicates the leading strand.

- **Xeroderma Pigmentosum:** Defect in **nucleotide excision repair (NER)** (UV-specific excinuclease/endonuclease), preventing removal of UV-induced pyrimidine (thymine-thymine) dimers and resulting in extreme photosensitivity and early skin malignancies.
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- **Lynch Syndrome (HNPCC):** Caused by mutations in genes governing **mismatch repair (MMR)** (e.g., *MSH2*, *MLH1*), resulting in microsatellite instability.
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Transcription, Post-Transcriptional Modification, and Disorders

- **Eukaryotic RNA Polymerase roles:** RNA Pol I synthesizes 28S, 18S, and 5.8S rRNA; **RNA Pol II synthesizes mRNA and snRNA**; RNA Pol III synthesizes tRNA and 5S rRNA.
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- **Toxin inhibition: -Amanitin** (from *Amanita phalloides* / death cap mushroom) selectively inhibits eukaryotic **RNA Polymerase II** at very low concentrations, causing fatal hepatic necrosis.
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- **Three mandatory mRNA modifications:** Addition of a 7-methylguanosine cap at the 5' end (via 5'-to-5' triphosphate linkage), 3' polyadenylation (cleavage followed by poly-A tail addition without a template), and removal of introns via the spliceosome.
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- **Splice-site defect disorder:** Mutations disrupting conserved 5' (GU) or 3' (AG) splice junctions cause aberrant splicing, frequently seen in forms of **-thalassemia**.
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Translation and Post-Translational Modification

- **Ribosomal decoding:** Wobble hypothesis permits non-Watson-Crick base pairing between the 3rd base (3' end) of the mRNA codon and the 1st base (5' end) of the tRNA anticodon.
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- **Antibiotic discrimination: Aminoglycosides** bind the 30S ribosomal subunit to cause mRNA misreading; **chloramphenicol** inhibits bacterial 50S peptidyltransferase; **erythromycin (macrolides)** halts translocation along the 50S subunit.
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- **Eukaryotic translation toxin: Diphtheria toxin** and *Pseudomonas* Exotoxin A inactivate elongation factor 2 (eEF-2) via NAD-dependent ADP-ribosylation, halting peptide elongation.
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- **Post-translational collagen modification:** Hydroxylation of proline and lysine residues occurs within the rough ER, strictly requiring **ascorbic acid (Vitamin C)**, , and -ketoglutarate; absence results in unhydroxylated collagen failing to form a stable triple helix (scurvy).
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Protein Synthesis and Gene Expression

- **Genetic code characteristics:** The code is degenerate (redundant), non-overlapping, unambiguous, universal, and read continuously without punctuation.
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- **Stop codons:** Standard eukaryotic termination codons are **UAA, UAG, and UGA**; they bind release factors rather than tRNAs.
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- **Epigenetic control:** DNA methylation at CpG islands universally silences/represses transcription, whereas histone acetylation relaxes chromatin (euchromatin) and activates gene expression.
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- **Operon dynamics (Prokaryotes):** In the *lac* operon, allolactose (inducer) releases the repressor from the operator; cyclic AMP (cAMP) bound to CAP/CRP promotes transcription when glucose is exhausted.
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Inborn Errors of Nucleic Acid Metabolism

- **Orotic Aciduria:** Autosomal recessive defect in **uridine monophosphate (UMP) synthase** (orotate phosphoribosyltransferase and OMP decarboxylase), presenting with megaloblastic anemia refractory to /folate, physical/mental retardation, and massive urinary orotic acid without hyperammonemia.
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- **Distinguishing OTC from UMP deficiency:** Ornithine transcarbamoylase (OTC) deficiency displays hyperammonemia, encephalopathy, and high orotic acid; UMP synthase deficiency causes isolated orotic aciduria and megaloblastic anemia with completely normal blood ammonia.
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- **Adenosine deaminase deficiency:** Primary metabolic defect producing extreme elevations in dATP, universally abolishing cell-mediated and humoral immunity (SCID).
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Protein Digestion, Absorption, and Gluconeogenesis

- **Zymogen cascade trigger:** Enteropeptidase (enterokinase) anchored on the duodenal brush border cleaves trypsinogen to active **trypsin**, which subsequently cleaves and activates chymotrypsinogen, proelastase, and procarboxypeptidases.
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- **Hartnup disease:** Defect in the neutral amino acid transporter in intestinal enterocytes and renal proximal tubules, leading to severe urinary loss of tryptophan and pellagra-like symptoms (dermatitis, diarrhea, dementia).
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- **Obligate gluconeogenic bypasses:** Pyruvate carboxylase (mitochondrial, requires biotin and obligate activator **acetyl-CoA**), PEPCK (requires GTP), Fructose 1,6-bisphosphatase-1, and Glucose 6-phosphatase (ER-bound, absent in muscle).
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- **Substrate flux:** Alanine serves as the principal gluconeogenic amino acid delivered from muscle to liver during fasting via the **glucose-alanine cycle**.
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Transamination, Deamination, Urea Cycle, and Toxicity

Coenzyme dependency: All transaminases (ALT, AST) require **pyridoxal phosphate (PLP / Vitamin B6)** as an obligatory intermediate carrier of amino groups.

Exclusive oxidative deamination: Hepatic **glutamate dehydrogenase** catalyzes the reversible oxidative deamination of glutamate to -ketoglutarate, uniquely utilizing either or as cofactors.

Ammonia trapping in brain: Excess cerebral ammonia depletes -ketoglutarate (inhibiting the TCA cycle) by driving glutamate and glutamine synthesis, leading to brain edema via glutamine-induced osmotic swelling in astrocytes.

Urea cycle stoichiometry: Urea incorporates two nitrogen atoms: the first originates from **free ammonium ()** via CPS-I, and the second is directly donated by **aspartate** via argininosuccinate synthetase.

Urea Cycle Disorders and Hyperammonemia

- **CPS-I vs. NAGS deficiency:** Carbamoyl phosphate synthetase I requires **N-acetylglutamate (NAG)** as an obligate allosteric activator; deficiency in either CPS-I or NAGS results in severe hyperammonemia with low blood citrulline and **absent urinary orotic acid**.
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- **Ornithine Transcarbamoylase (OTC) deficiency:** The most common urea cycle disorder, inherited as **X-linked recessive** (all others are autosomal recessive), presenting with profound hyperammonemia, low citrulline, and **markedly elevated urinary orotic acid**.
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- **Emergency pharmacotherapy:** Hyperammonemia is treated with intravenous **sodium phenylbutyrate** (or phenylacetate, conjugating glutamine) and **sodium benzoate** (conjugating glycine) to bypass the compromised urea cycle for nitrogen excretion.
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Metabolism and Inborn Errors of Tryptophan & Others

- **Tryptophan derivatives:** Serves as the biochemical precursor for **serotonin**, **melatonin**, and **niacin** (dietary tryptophan yields nicotinic acid).
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- **Carcinoid Syndrome:** Neuroendocrine tumors overproduce serotonin, consuming massive amounts of systemic tryptophan and triggering secondary niacin deficiency (pellagra: diarrhea, dermatitis, dementia) alongside elevated urinary **5-HIAA**.
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- **Maple Syrup Urine Disease (MSUD):** Deficiency in the **branched-chain -keto acid dehydrogenase complex (BCKDC)**, preventing catabolism of leucine, isoleucine, and valine; toxic ketoacids yield sweet maple-syrup-smelling urine and severe neurotoxicity.
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Metabolism of Phenylalanine and Tyrosine

- **Classical Phenylketonuria (PKU):** Deficiency of **phenylalanine hydroxylase (PAH)**, rendering tyrosine an essential amino acid, leading to elevated phenylketones (phenylacetate, phenyllactate) in urine causing a characteristic "mousy" or "musty" odor.

- **Malignant/Atypical PKU:** Caused by defective **dihydrobiopterin reductase (DHPR)** or biopterin synthesis; diet restriction of phenylalanine fails to rescue neurotransmitter production (dopamine, serotonin).
- **Alkaptonuria:** Autosomal recessive deficiency of **homogentisate oxidase** in the tyrosine degradation pathway; features urine that turns jet black upon standing/alkalinization, ochronosis (dark cartilage pigment), and debilitating arthropathies.
- **Albinism:** Autosomal recessive deficiency of **tyrosinase** (copper-containing enzyme), blocking the conversion of DOPA into melanin pigments.

Specialized Amino Acid Metabolism and Inborn Errors

- **Homocystinuria:** Classical type caused by **cystathionine -synthase** deficiency (requires Vitamin B6); manifests with ectopia lentis (downward/inward lens subluxation), marfanoid habitus, osteoporosis, and fatal premature thromboembolism.
- **Distinction from Marfan syndrome:** Ectopia lentis subluxates **downward and inward** in homocystinuria (with cognitive impairment and thromboses), whereas it subluxates **upward and outward** in Marfan syndrome (fibrillin-1 defect, normal intellect).
- **Creatine synthesis localization:** Initiated in kidneys (arginine + glycine guanidinoacetate) and finalized in liver (SAM-dependent methylation to creatine) before transport to muscle and brain.
- **Nitric oxide generation:** Synthesized from **L-arginine**, , and NADPH by nitric oxide synthase (NOS), yielding nitric oxide and citrulline.

Introduction and Classification of Neurotransmitters

- **Classification scheme:** Biogenic amines (acetylcholine, catecholamines, serotonin, histamine), amino acids (glutamate, aspartate = excitatory; GABA, glycine = inhibitory), and purines/neuropeptides.

- **GABA synthesis:** Formed by irreversible -decarboxylation of L-glutamate catalyzed by **glutamate decarboxylase (GAD)**, which strictly requires **pyridoxal phosphate (PLP / Vitamin B6)**.
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- **Catecholamine cascade:** Phenylalanine Tyrosine L-DOPA Dopamine Norepinephrine Epinephrine; the rate-limiting enzyme is **tyrosine hydroxylase** (requires).
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- **Epinephrine methylation:** Phenylethanolamine N-methyltransferase (PNMT) converts norepinephrine to epinephrine in the adrenal medulla, using **S-adenosylmethionine (SAM)** as methyl donor, strongly induced by cortisol.
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Major Neurotransmitters in Oral Function

- **Salivary secretomotor drive:** Parasympathetic postganglionic release of **acetylcholine (ACh)** binds muscarinic receptors on acinar cells, triggering a massive signaling cascade that produces watery, amylase-rich saliva.
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- **Protein-rich salivary secretion:** Sympathetic postganglionic release of **norepinephrine** acts on -adrenergic receptors, increasing intracellular cAMP and driving exocytosis of salivary mucins and proline-rich proteins.
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- **Oral nociception and neurogenic inflammation:** Primary sensory afferents (trigeminal nerve / C-fibers) release **Substance P** and **Calcitonin Gene-Related Peptide (CGRP)** in dental pulp and periodontium upon noxious stimulation, inducing local vasodilation, mast cell degranulation, and plasma extravasation.
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Metabolic Pathways: Glycolysis, TCA Cycle, and Protein Metabolism in Starvation and Exercise

Anaplerosis: Pyruvate carboxylase replenishes oxaloacetate directly from pyruvate to maintain the TCA cycle when oxaloacetate is siphoned off for gluconeogenesis.

Metabolic triage in starvation (Day 1 to 3): Liver glycogen depletes within 24 hours; muscle protein proteolysis surges to supply amino acids (primarily alanine and glutamine) for hepatic gluconeogenesis.

Metabolic triage in prolonged starvation (> 3 days to weeks): Brain adapts to oxidize **ketone bodies (acetoacetate and -hydroxybutyrate)**, substantially sparing skeletal muscle proteolysis and blunting urea excretion.

Metabolism in vigorous exercise: Muscle relies on glycogenolysis and anaerobic glycolysis; high NADH/ ratio shunts pyruvate to lactate, which enters the bloodstream to enter the **Cori cycle** in the liver for resynthesis of glucose.

Bioenergetics: ETC and Disorders of Oxidative Phosphorylation

- **Complex sequence and electron convergence:** Electrons from Complex I () and Complex II (, containing FAD) converge at mobile carrier **coenzyme Q (ubiquinone)**, flowing to Complex III, cytochrome c, Complex IV, and finally reducing to .
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- **Specific ETC Inhibitors:** Rotenone and Amytal block Complex I; Antimycin A blocks Complex III; **Cyanide () and Carbon Monoxide (CO)** bind the ferric/ferrous iron in Complex IV (cytochrome), totally arresting mitochondrial respiration.
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- **Uncouplers vs. ATP Synthase Inhibitors: Oligomycin** binds the domain of Complex V, directly blocking the proton channel and secondarily halting the ETC; chemical uncouplers like **2,4-dinitrophenol (DNP)** and endogenous **thermogenin (UCP-1)** dissipate the proton gradient, collapsing ATP synthesis while stimulating oxygen consumption and releasing energy purely as heat.
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- **Leber Hereditary Optic Neuropathy (LHON):** Mitochondrial DNA point mutations (most frequently in Complex I genes) causing bilateral painless central vision loss in young adults.
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Role of Free Radicals and Oxidative Stress in Oral Diseases

Fenton and Haber-Weiss reactions: Hydrogen peroxide () interacts with ferrous iron () via the Fenton reaction to generate the **hydroxyl radical ($\cdot\text{OH}$)**, the most reactive and damaging radical in biological systems.

Periodontal tissue destruction: Neutrophil respiratory burst (via membrane-bound **NADPH oxidase** generating superoxide) releases excess reactive oxygen species (ROS) into periodontal pockets, upregulating host Matrix Metalloproteinases (MMPs like MMP-8/collagenase) and triggering alveolar bone resorption through RANKL activation.

Oral premalignancy and leukoplakia: Chronic tobacco/areca nut exposure generates persistent intracellular oxidative stress, driving lipid peroxidation, producing 8-hydroxy-2'-deoxyguanosine (8-OHdG) DNA adducts, and mutating the *TP53* tumor suppressor gene.

Vitamins and Antioxidants in Relation to Oral Health / Oral Diseases

Vitamin C (Ascorbic acid): Cofactor for prolyl and lysyl hydroxylases; severe deficiency (scurvy) results in defectively crosslinked collagen manifesting as fragile periodontal ligaments, loose teeth, swollen bleeding gingiva, and impaired oral mucosal wound healing.

Enzymatic antioxidant trio: Superoxide Dismutase (SOD) dismutates into ; **Catalase** converts into and ; **Glutathione Peroxidase** (a **selenium-dependent** enzyme) neutralizes peroxides using reduced glutathione (GSH).

Vitamin A (Retinoids): Preserves mucosal epithelial differentiation; deficiency causes squamous metaplasia, hyperkeratinization of oral mucosal membranes, and salivary gland atrophy leading to severe xerostomia.

Vitamin E (-tocopherol): Potent chain-breaking, lipid-soluble antioxidant located inside cell membranes, preventing free radical propagation during lipid peroxidation of periodontal tissue membrane polyunsaturated fatty acids.